

Emmanuel Maduabum

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ELECTRICAL ENGINEER · DATA ANALYSIS & AUTOMATION

Electrical and electronic engineer with commercial project experience across Ghana and Nigeria and a master's in intelligent manufacturing and control engineering in China. I design and audit energy systems, build the financial models and technical documentation that decisions get made on, and work in Python, MATLAB, Power BI and Excel. Currently running logistics, budgeting and compliance for a trading business in Nigeria while completing my MEng. Practical, detail-driven, and comfortable owning a problem from site survey through to the spreadsheet the client signs off.

KEY COMPETENCIES

Data Analysis & Power BI	Financial Modelling	Energy Auditing
Python & MATLAB	PVsyst & AutoCAD 2D/3D	Renewable Energy Modelling
Technical Documentation	Project Coordination	Robotics & Control

PROFESSIONAL EXPERIENCE

Business Operations Manager

Oct 2023 – Present

Cotto Enterprise · Abia State, Nigeria · petroleum lubricants trading

Managing end-to-end sales, logistics and financial operations for a commodities business with over ₦200 million in annual turnover. Overseeing product distribution across multiple regions, maintaining client relationships, and coordinating deliveries against targets and deadlines.

Accomplishments

- Coordinated multi-region distribution and logistics, liaising with suppliers, transporters and customers.
- Managed budgeting, financial administration and regulatory-compliance documentation.
- Oversaw sales and inventory operations across regions, resolving scheduling and delivery constraints under time pressure.
- Handled customer enquiries and maintained ongoing client relationships.

Project Engineer

Mar 2023 – Dec 2023

Mcjones Energy · Accra, Ghana

Planning, coordinating and monitoring engineering projects from concept to completion. Performed energy audits, feasibility studies and renewable energy simulations in PVsyst, and prepared the technical reports, financial models and project documentation supporting implementation decisions.

Accomplishments

- Planned, scheduled and monitored assigned engineering projects to timeline and quality standards.
- Performed energy audits assessing consumption patterns and identifying cost and energy savings.
- Prepared equipment specifications, feasibility studies and client-facing documentation in Excel and Word.
- Developed financial models with sensitivity analysis and income projections for project evaluation.

Junior Research Analyst

Sep 2022 – Feb 2023

HerrnHut Corp · Tema, Ghana

Supported data collection, analysis and interpretation to guide business and project decisions. Conducted market and technical research, organised findings into clear reports, and contributed to the financial modelling behind a multi-site eco-centre development.

Accomplishments

- Contributed to successive versions of a consolidated project financial model for the Botto Eco-Centre development, including pre-tax and after-tax scenario variants.
- Produced a site-by-site power requirement analysis used to size the development's energy provision.
- Developed project strategy plans using logical framework approaches; maintained timeframes, objectives and communications.
- Tracked, forecast and reported on project progress including metrics and challenges.

Electrical Engineering Intern

Sep 2021 – Feb 2022

GFEEL Engineering · Tema, Ghana

Assisted in the design, installation and maintenance of electrical systems for residential, commercial and industrial projects — wiring, CCTV, solar and electrical fence installations — supporting senior engineers with technical drawings, cost estimates and field inspections.

PROJECTS

Autonomous Lawn Mower Robot — *self-built platform · hardware complete, autonomy in development*

- Designed, built and wired an outdoor autonomous mowing platform — chassis, drive system, power and control electronics — and maintain it as a working prototype.
- Developing the navigation stack: ROS with RTAB-Map for mapping and localisation, and evaluating reinforcement learning for coverage path planning.
- Working through the failure modes that only appear outdoors: uneven terrain, variable lighting and seasonal change in the environment.

Green Hydrogen & Fuel Cell Feasibility Study — *technical and economic study*

- Study of coupling an electrolyser and fuel cell to surplus renewable and waste-to-energy generation to store and firm intermittent power.
- Covers technology selection, indicative sizing against available surplus power and water, and the conditions under which a demonstration unit becomes economically sensible before scaling.

Solar Irradiance Forecasting with Deep Learning — *MATLAB · LSTM, ConvLSTM · time-series modelling*

- Built and evaluated recurrent and convolutional-recurrent models for short-horizon solar irradiance prediction, comparing architectures on forecast accuracy.

Cloud Classification with Convolutional Neural Networks — *MATLAB · CNN · image processing*

- Trained a CNN to classify sky-image cloud types — the perception half of a solar nowcasting system, and a companion to the forecasting work above.

Sales Performance Dashboard — *Microsoft Power BI · Dataleum team project*

- Built an interactive sales dashboard from a raw transactional dataset: data cleaning and shaping, calculated measures, and visualisations designed for a non-technical audience.

Speed-Bump Energy Generator for Street Lighting — *mechanical design · energy conversion · system modelling*

- Designed a road-mounted energy harvester converting vehicle motion into electricity for street lighting, including the conversion mechanism and system-level modelling.

EDUCATION

Master of Engineering (MEng), Intelligent Manufacturing and Control Engineering

Sep 2025 – Jan 2028

Zhejiang University of Science and Technology · Hangzhou, Zhejiang, China

- Awarded the Scholarship for Outstanding New Students (50% of first-year tuition).
- Research area: probabilistic solar irradiance forecasting and autonomous systems.

Bachelor of Engineering (BEng), Electrical and Electronic Engineering

2019 – 2023

Accra Institute of Technology · Accra, Ghana

CERTIFICATES

- Data Analytics and Business Intelligence — Dataleum (January 2024)
- Project Development for Renewable Energies — Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH (January 2023)
- Duolingo English Test: 120, CEFR B2 (December 2024)

LANGUAGES & LEADERSHIP

English (fluent) · Igbo (native) · Mandarin Chinese (HSK 2, improving)

- Minister, formerly Captain, Sports & Arts Department, ZUST International Student Union — led the university team to first place at the 8th Water Sports Competition.

REFERENCE

Mr. Jones Elias Ntorinkansah

Managing Director & CEO, McJones Energy, Accra, Ghana

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